

Photometric measurements of the dissociation of manganese oxalate

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Abstract

In this experiment, the absorbance of a solution consisting of manganese sulfate, oxalic acid and potassium permanganate is measured for several times after the substances were combined. This series of measurements is done for the temperatures 23.2(5) °C, 26.3(5) °C and 35.1(5) °C. For each temperature, the rate constant k is determined, the results are $1.486(5) \times 10^{-3} \text{ s}^{-1}$, $1.814(6) \times 10^{-3} \text{ s}^{-1}$ and $6.31(4) \times 10^{-3} \text{ s}^{-1}$, respectively. From these three values, the frequency factor $A = 3.4(3) \times 10^8 \text{ s}^{-1}$ and the activation energy $E_a = 6.362(12) \times 10^4 \text{ J/mole}$ are determined by applying Arrhenius equation.

reaction?

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0 Introduction

The reaction rate determines, how fast a chemical reaction takes place under certain conditions. It is mainly described by the concentration of the reactants, but also the activation energy plays a crucial role. In industrial applications, for instance in the manufacturing of chemicals like ammonia, but also in the production of pharmaceuticals, some reactions have to take place. Thus, the reaction rate is an important indicator of the speed at which a reaction occurs.

This importance can be seen as a motivation of this lab report. A complex reaction between manganese sulfate, oxalic acid and potassium permanganate are investigated, by exposing the solution to light of 520 nm and measuring the absorbance at several times. There are three series of measurements done for three temperatures. For each temperature, the reaction constant is determined. In the end, the frequency factor and the activation energy of the analyzed reaction are computed by Arrhenius equation.

The report starts with a brief theoretical input, then describes the experimental setup. Then, the results are presented, followed by a discussion.

1 Theory

In this section, the most relevant theory is presented, including information about reaction rates, the Arrhenius equation and the chemical reaction that is investigated. If not stated otherwise, the content refers to the script [1].

1.1 Beer–Lambert law

If light passes a medium, the intensity decreases exponentially with the transversed distance d , which is described by the Beer-Lambert law, which states

$$I = I_0 10^{-\epsilon \cdot c \cdot d}, \quad (1) \quad \checkmark$$

where I is the intensity after a certain distance d , ϵ the molar absorption coefficient and c the molar concentration. The transmittance T is then defined as I/I_0 and describes the fraction of intensity left after passing a medium.

1.2 Reaction rates

The velocity of an occurring reaction is determined by its reaction rate, which is defined as the change of concentration divided by the time difference. The general form of the reaction rate law for a reaction



where the capital letters are the reactants and products, and the lower case letters the prefactors, can be formulated as

$$\nu = k[A]^m[B]^n. \quad (3)$$

Thereby, the rate ν depends on the rate constant k , which is depending on temperature, the concentrations $[A]$ and $[B]$ of the respective reactant, and the factors m and n . The factors m and n define the order of the reaction, which is the sum of both, and can in general only be determined experimentally. For special reactions, this law has a simplified form. For first order reactions, for instance in the form $A \rightarrow B + C$, the concentration rate is depending linearly on the concentration of A :

$$\nu = -\frac{d[A]}{dt} = k[A]. \quad (4)$$

Solving this differential equation yields

$$[A] = [A]_0 e^{-k \cdot t}. \quad (5)$$

For second order equations of the type $A + B \rightarrow P$, the relation gets more complicated, the rate depends linearly on the two reactants concentrations:

$$\nu = -\frac{d[A]}{dt} = k[A][B]. \quad (6)$$

The solution for this equation is much more complicated and requires the monitoring of two concentrations at each time. An approximation can be done by adding a very high concentration

of one reactant. Thus, its concentration remains nearly constant, and the rate only depends linearly on the other reactant's concentration:

$$\nu = k'[B], \quad (7)$$

where $k' = k[A]$, and the concentration $[B]$ is much lower than $[A]$.

1.3 Arrhenius equation

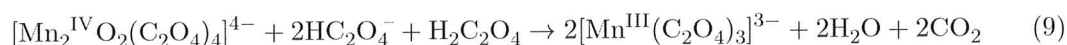
The Arrhenius equation provides a connection between the rate constant k and the activation energy E_a of a reaction, which is the energy barrier that has to be overcome for the reaction to occur. The equation states

$$k = Ae^{-\frac{E_a}{RT}}, \quad (8)$$

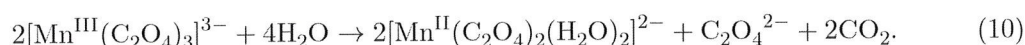
where R is the gas constant, T the temperature and A is called the frequency factor. This factor provides information about the number of collisions between reactant molecules.

1.4 Reaction

In the reaction that is studied in the experiment [2], manganese sulfate MnSO_4 , oxalic acid $\text{C}_2\text{H}_2\text{O}_4$ and potassium permanganate KMnO_4 are reacting. The reaction occurs in different steps and contains a redox reaction where the oxidation state of manganese changes from +4 to +3, as it gains one additional electron, donated by oxalic acid. The first step of the reaction reads



Then, the second step takes place:



Thus, in the end, manganese gets reduced from Mn^{IV} gets reduced to Mn^{II} . The reaction rate can be written as

$$\nu = \alpha[\text{Mn}^{\text{III}}] + \beta[\text{Mn}_2^{\text{IV}}]^{1/2}[\text{Mn}^{\text{III}}] + \gamma[\text{Mn}_2^{\text{IV}}]^{1/2}, \quad (11)$$

where α , β and γ are the reaction constants for the respective substance. As the concentration of Mn_2^{IV} remains nearly constant, the value of α is the best value to estimate the rate of our experiment. In [2], four values of α are given for four temperatures, as shown in Table 1. Thus, in Figure 3, the here given values for α are used to create a fit to determine the literature value of the frequency factor and the activation energy.

Table 1: Literature values for α for four temperatures, values taken from [2].

T ($^{\circ}\text{C}$)	α (min^{-1})
10	$2.39 \cdot 10^{-3}$
20	$0.85 \cdot 10^{-2}$
30	$3.00 \cdot 10^{-2}$
40	$0.85 \cdot 10^{-1}$

2 Setup and procedure

In the experiment, three solutions are used: Manganese sulfate MnSO_4 at a concentration of 0.2 mole/l, oxalic acid $\text{C}_2\text{H}_2\text{O}_4$ at the same concentration, and potassium permanganate KMnO_4 at 0.02 mole/l.

The apparatus to measure the intensity of light after passing a probe is as follows: A mercury gas lamp is utilized. The light is filtered with a 520 nm filter located in front of the lamp. After the filter there is a mount, in which the probe solution can be placed. After the cell a Polytec i-trometer is measuring the intensity of the light transmitted from the cell.

At first, a spectrum of each of the solutions and of distilled water is measured for reference. Then, 2 ml of MnSO_4 are mixed with 14 ml of the oxalic acid solution. Then, the flask is shaken and it has to be waited for five minutes. Afterwards, 4 ml of the KMnO_4 solution are added with the pipette. The time of this insertion has to be noted. Immediately, approximately 2.5 ml of the mixture are inserted into the cell for measuring the lamp's intensity. The first series of measurements is accomplished at room temperature, $23.2(5) ^{\circ}\text{C}$. Every 30 seconds a spectrum is measured, until no significant change between the single measurements is recognized any more.

The second and third series of measurements are done at other temperatures, at $26.3(5) ^{\circ}\text{C}$ and $35.1(5) ^{\circ}\text{C}$. For every series, a new mixture has to be done the same way as described before. To receive the different temperatures, the mixture has to be kept in a temperature-regulated water bath. Between two measurements, the liquid has to be returned to the rest in the water bath with a pipette, to keep the temperature approximately constant.

3 Results

We start the experiment by recording a spectrum of distilled water to get a calibration measurement for the intensity. The recorded spectrum can be seen on the left in Figure 1 and shows a pronounced peak at ≈ 520 nm corresponding to an emission line of the mercury lamp. To extract the peak height we identify the single highest datapoint in the spectrum and fit a parabola of the form $f(x) = a(x - b)^2 + c$ to datapoints. We tried fitting Gaussian or Lorentz profiles to the full peaks but neither accurately captured the roundness on the top of the peak. Since we are only interested in the peak height fitting a parabola to the topmost datapoints allows us to estimate the peak height as well as uncertainty.

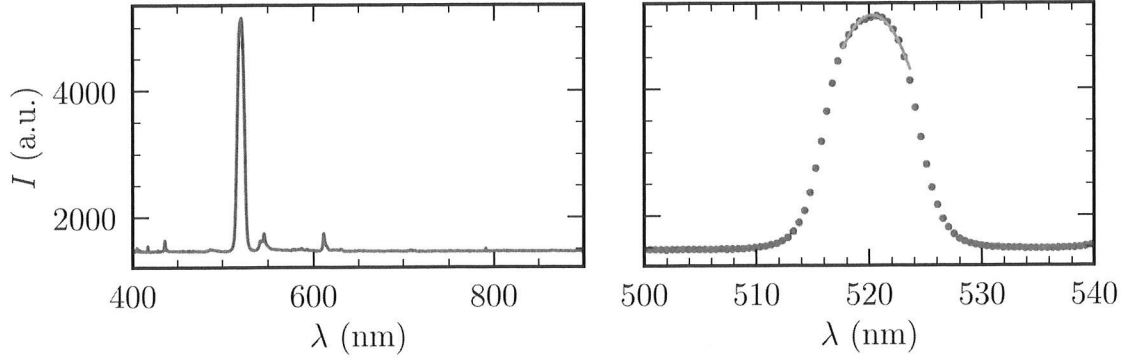


Figure 1: In the left plot the recorded spectrum of distilled water can be seen. A sharp peak at ≈ 520 nm corresponding to an emission line of the mercury lamp can be seen. On the right a close-up view of the peak is shown. A parabola was fitted to the 10 highest datapoints and can be seen as solid orange line.

We use the peak height of distilled water $I_0 = 5168(15)$ as reference to normalize the following measurements. We proceed by performing intensity measurements every 30 s for temperatures $23.2(5)^\circ\text{C}$, $26.3(5)^\circ\text{C}$ and $35.1(5)^\circ\text{C}$. The collected data can be seen in Figure 2. Functions of the form $f(x) = a \exp(-kt)$ were fitted to the data and can be seen as solid lines. The first few datapoints (enclosed by the red dashed box) deviate significant from the exponential trend and are therefore excluded from the fit. The reason for this behavior is presumably the temperature stabilization, which does not warm the probe to the wanted temperature instantly.

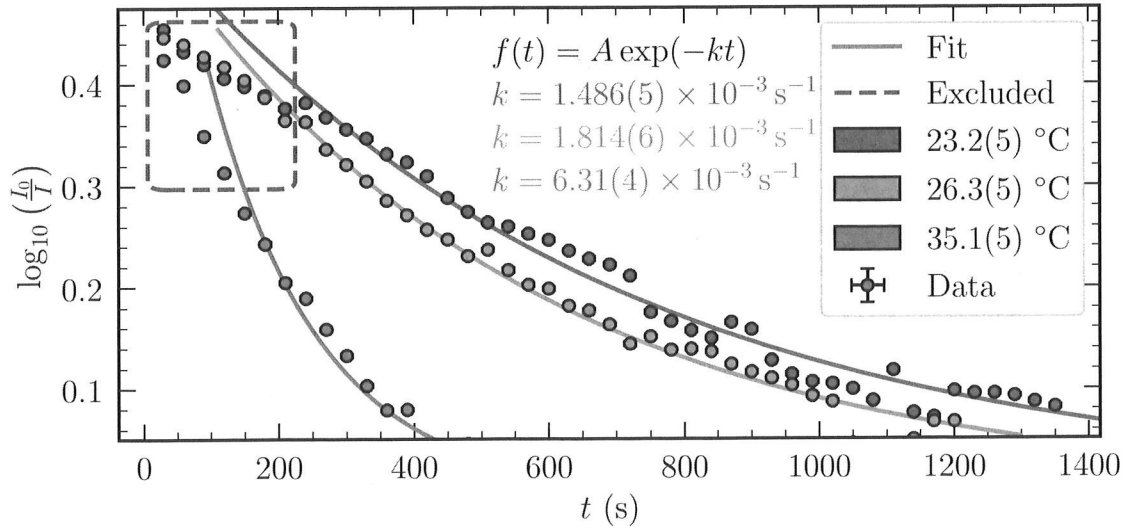


Figure 2: The decadic logarithm of the reference intensity divided by the intensity is plotted for each measurement, dependent on time. Three series of measurements are shown color-coded for the three temperatures. The data points in the dotted red box are excluded, as they clearly do not follow the exponential regime that was used for fitting. The uncertainties are too small to be made out.

The reaction rates determined from the fits are summarized in Table 2. The rate constant k is plotted against the respective temperature in Figure 3. An exponential fit based on the

Arrhenius equation (see Equation (8)) is applied, in order to determine the frequency factor A and the activation energy E_a of the reaction. Literature values of k are plotted, alongside an equivalent fit. The literature values are introduced in Table 1. The fit for the literature values yields expectation values for the frequency factor and activation energy as well as the possibility to determine a rate constant k for every temperature, although only four values were given.

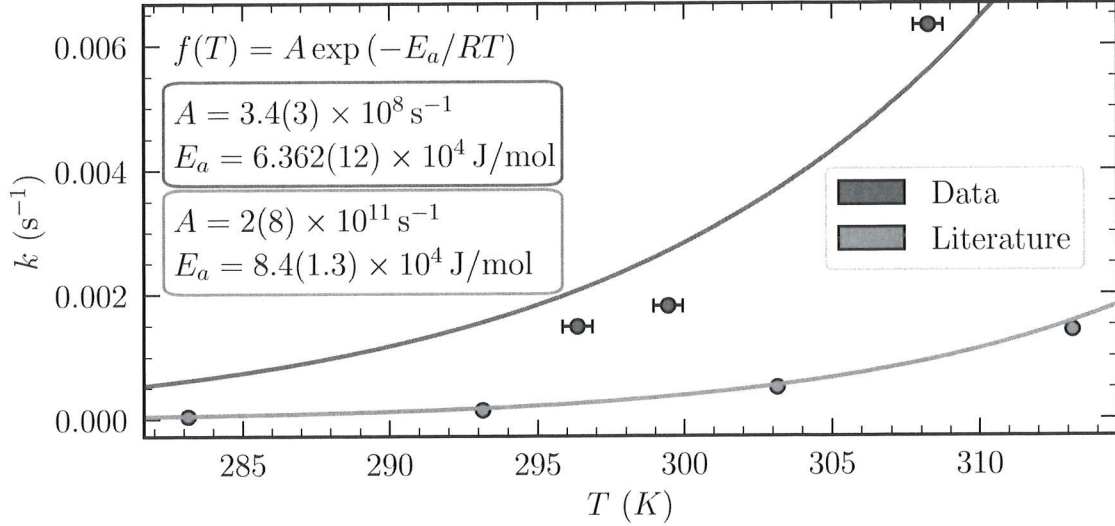


Figure 3: The measured values of the rate constant k are plotted against the temperature, alongside an exponential fit. Additionally, literature values from [2] are plotted and fitted, to create a reference.

Using the model fitted to the literature values we can use interpolation to predict the rate constants at the temperatures we measured at. These values are shown in Table 2 alongside the measured rate constants. Due to the large uncertainty in the literature fit (especially in A) we do not get values which we can reasonably compare. Especially if we notice that the uncertainty is larger than the nominal value, at which point Gaussian error propagation breaks down. The large fit uncertainty is possibly an artifact of few datapoints, combined with the fact that we fit for large values ($\sim 10^{11}$) which might push the numerical minimization algorithm to its limits.

Table 2: Measured and literature values for k for three temperatures.

T ($^{\circ}\text{C}$)	k_{measured} (s^{-1})	k_{lit} (s^{-1})
23.2(5)	$1.486(5) \cdot 10^{-3}$	$3(21) \cdot 10^{-4}$
26.3(5)	$1.814(6) \cdot 10^{-3}$	$4(29) \cdot 10^{-4}$
35.1(5)	$6.31(4) \cdot 10^{-3}$	$1(3) \cdot 10^{-3}$

Nevertheless we can compare the fitted frequency factor and activation energy extracted from our measurements

$$A = 4.3(3) \times 10^8 \text{ s}^{-1} \quad \text{and} \quad E_a = 6.362(12) \times 10^4 \text{ J/mol} \quad (12)$$

with the ones from the literature fit

$$A_{\text{lit}} = 2(8) \times 10^{11} \text{ s}^{-1} \quad \text{and} \quad E_{a,\text{lit}} = 8.4(1.3) \times 10^4 \text{ J/mol.} \quad (13)$$

The literature frequency factor can not meaningfully be compared due to the large uncertainty, we can however compare the activation energies. We are within 2σ from the literature value lying in the gray zone between statistically likely and almost impossible. More work is therefore needed before more meaningful results can be produced. ✓

4 Discussion

In this experiment we successfully used photometric measurements to measure the dissociation of manganese sulfate in a complex reaction. By measuring the absorbance at regular time intervals of 30 s and fitting an exponential decay function we extract the rate constant k for the chemical reaction at three different temperatures $k_{23.2(5) \text{ } ^\circ\text{C}} = 1.486(5) \times 10^{-3} \text{ s}^{-1}$, $k_{26.3(5) \text{ } ^\circ\text{C}} = 1.814(6) \times 10^{-3} \text{ s}^{-1}$ and $k_{35.1(5) \text{ } ^\circ\text{C}} = 6.311(4) \times 10^{-3} \text{ s}^{-1}$.

We proceed by fitting the Arrhenius equation to the reaction rates to extract the frequency factor $A = 4.3(3) \times 10^8 \text{ s}^{-1}$ and activation energy $E_a = 6.362(12) \times 10^4 \text{ J/mol}$. By also fitting to literature values extracted from [2] we can compare both values. The frequency factor can not be meaningfully compared due to the high uncertainty in the fit parameters of the literature fit whilst the activation energy is within 2σ . To increase the statistical significance of our results a few things can be pointed out. By having to transfer the probe between heat bath and measurement setup we introduce systematic errors into our measurements. A possible improvement consists of being able to heat the probe in the measurement setup. Additionally measuring the rate constant for more temperatures increases the degrees of freedom of the Arrhenius fit, leading to a more accurate determination of frequency rate and activation energy. ✓

5 Questions

a) Why is potassium permanganate violet?

The reason for the violet color of potassium permanganate lies in the light absorbing and emitting properties. The main absorption occurs in the yellow-green region of wavelengths. The remaining light that is reflected and transmitted appears purple. The intense absorption is mainly a result of charge-transfer reactions, in which electrons are transferred from the oxygen atom to empty d-orbitals in the Mn atom. ✓

b) Explain the occurrence of an absorption line by the Franck-Condon principle (Not more than one page).

The Franck-Condon principle describes absorption lines of vibrational transitions or emission or absorption of photons. The line intensity thereby depends on the overlap of the wave functions of the two states that are involved in a transition. In general the Franck-Condon principle is a result of the Born-Oppenheimer approximation, in which the movement of the atomic nuclei and the electronic states are separated. As the internuclear distance changes on a much larger

timescale than the electronic states. Thus, the distance between atoms remains constant while the absorption or emission of a photon occurs. This is the reason why transitions only occur vertically in the potential-distance diagram. Thus, only transitions with an overlapping wave function are allowed. The higher the overlap, the higher the transition intensity.

c) A piece of wood found during an archaeological dig was analyzed using the ^{14}C method. Comparison to a living tree showed that the concentration of ^{14}C was 28 % lower. Determine the rate constant k and how old the piece of wood is. The half-life of ^{14}C is 5730 a.

As this process is a radioactive decay, it is a first-order reaction. Thus, the formula

$$t_{1/2} = \frac{\ln 2}{k} \Rightarrow k = \frac{\ln 2}{t_{1/2}}. \quad (14)$$

Inserting the half-life yields $k = 1.20968 \times 10^{-4} \text{ years}^{-1}$. From the relation

$$[A] = [A]_0 e^{-k \cdot t} \quad (15)$$

t can be set free:

$$t = \ln \left(\frac{[A]}{[A]_0} \right) / k = 2715.63 \text{ years}. \quad (16)$$

The ratio $[A]/[A]_0$ equals 0.72.

d) During a lot of chemical experiments not only the main product is formed but also side-reactions to very different products can take place. Consider that upon heating reactant A can irreversibly react forming different products B or C (sketch seen below). Write down the equations for concentrations $[A]$, $[B]$ and $[C]$ of a first order reaction. How would a concentration vs. time diagram for the reaction look like for $k_C > k_B$.

If A can react either to A or B, the concentration behaves like

$$\frac{d[A]}{dt} = -(k_B + k_C)[A], \quad (17)$$

which yields

$$[A] = [A]_0 e^{-(k_B + k_C)t}. \quad (18)$$

For B and C the equations

$$\frac{d[B]}{dt} = (k_B)[A] \text{ and } \frac{d[C]}{dt} = (k_C)[A] \quad (19)$$

are valid. The time evolution of the concentrations of B and C can then be derived as

$$[B] = [A]_0 \frac{k_B}{k_B + k_C} (1 - e^{-(k_B + k_C)t}) \quad (20)$$

and equivalently

$$[C] = [A]_0 \frac{k_C}{k_B + k_C} (1 - e^{-(k_B + k_C)t}). \quad (21)$$

It contains the concentration of $[A]_0$ and its decrease, but also the branching ratio to B and C,

represented by k_B and k_C . The three concentrations are seen in Figure 4.

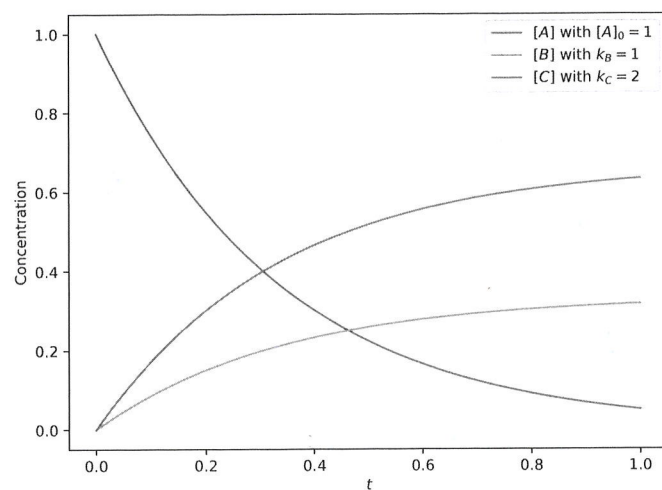


Figure 4: Concentrations of three substances with A reacting to either B or C, with the reaction constants $k_B = 1$ and $k_C = 2$. All units are arbitrarily chosen.



References

- [1] *Photometric measurements of the dissociation of manganese oxalate*. Universität Innsbruck, Innsbruck, AT, 2024.
- [2] T. Uehiro, I. Taminaga, and Y. Yoshino. "The Decomposition of the Oxalato Complex of Manganese(IV) in Oxalate Buffer Solutions". In: *Bulletin of the Chemical Society of Japan* 48.10 (Mar. 2006), pp. 2809–2812. ISSN: 0009-2673. eprint: <https://academic.oup.com/bcsj/article-pdf/48/10/2809/56091328/bcsj.48.2809.pdf>. URL: <https://doi.org/10.1246/bcsj.48.2809>.

Journal abbreviations:

Typical citation style.

T. Uehiro, I. Taminaga, Y. Yoshino, *Bull. Chem. Soc. Jpn.*
1975, 48, 2809–2812 ; DOI...

Erklärung

Hiermit versichern wir, dass der vorliegende Bericht selbständig verfasst wurde und alle notwendigen Quellen und Referenzen angegeben sind.



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Alexander Helbok

January 30, 2025

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Date



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January 30, 2025

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Date

